

KINGDOM OF ESWATINI
Ministry of Labour and Social Security
Measurement and Testing Unit

Software Requirements Specification

SDS Test System

Online Self-Directed Search Career Assessment Platform

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This revision incorporates TWG’s comments on business flow visibility and accuracy, production-readiness requirements, acceptance governance, ownership provisions, and document control.

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Version	Date	Summary of Change	Author / Editor
1.0	March 2026	Initial Draft	SDS Development Team
1.1	April 2026	Updated SRS to incorporate RSTP review comments: revised business flow diagrams; Kubernetes, IaC, and CI/CD requirements; UAT defect severity matrix; intellectual property clause; revised document control; corrected NFR-061 wording.	SDS Development Team

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional, non-functional, interface, operational, and acceptance requirements for the Online Self-Directed Search (SDS) Test System. The solution implements Holland's RIASEC career assessment model for the Ministry of Labour and Social Security of the Kingdom of Eswatini and establishes the baseline against which solution design, delivery, testing, and user acceptance shall be measured.

1.2 Document Scope

- Complete system requirements for the SDS Test System.
- Functional requirements for all supported user roles.
- Non-functional requirements covering security, performance, reliability, maintainability, and deployment readiness.
- System interfaces, constraints, assumptions, dependencies, and acceptance requirements.
- Required delivery artefacts needed for production-ready implementation.

1.3 Intended Audience

- Ministry of Labour and Social Security stakeholders.
- System developers, architects, and DevOps engineers.
- Quality assurance and user acceptance teams.
- Project managers, analysts, and implementation partners.
- End users, including test takers, counsellors, test administrators, and system administrators.

1.4 References

- Holland, J. L. (1997). Making Vocational Choices: A Theory of Vocational Personalities and Work Environments.
- Eswatini Data Protection Act 2022.
- WCAG 2.1 Accessibility Guidelines.
- OWASP Top 10 Security Guidelines.
- System Design Document (SDS Test System).

1.5 Business Model

The SDS Test System forms part of the Ministry's career guidance service ecosystem and is intended to modernize assessment delivery, recommendation generation, and policy-facing reporting across Eswatini.

1.5.1 Organizational Context

Entity	Role in the Ecosystem
Ministry of Labour and Social Security	Primary stakeholder and system owner.
Measurement and Testing Unit	Operational department is responsible for career guidance services and business ownership.
Educational Institutions	Partner schools, colleges, and universities that use assessment outputs.

Entity	Role in the Ecosystem
Employers and Industry Bodies	Beneficiaries of better career alignment and labour-market insight.
Students and Professionals	End users receive career guidance, recommendations, and certificates.

1.5.2 Business Functions

- Career Assessment: automated RIASEC-based personality and interest assessment.
- Career Guidance: personalized recommendations based on assessment results and local pathways.
- Data Analytics: labour-market insight and trend analysis to support policy planning.
- Institution Management: coordination with educational institutions and other approved stakeholders.
- Reporting and Compliance: auditable reporting, records management, and data protection compliance.

1.5.3 Current-State Business Flow (As-Is)

The current service model relies on manual scheduling, manual or facilitator-led administration, manual scoring, and delayed reporting. This makes it difficult to scale the service consistently and obscures the business problems the SDS platform must solve.

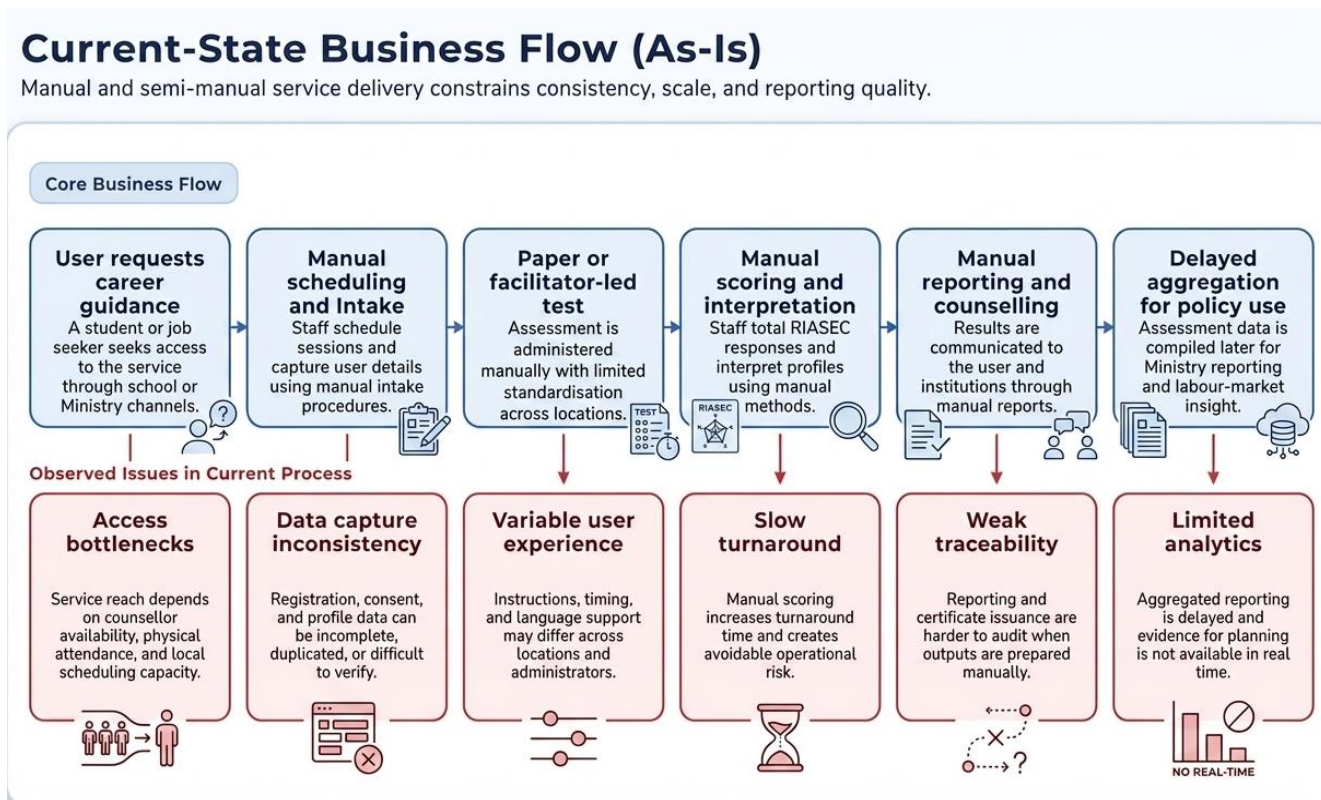


Figure 1. Current-state business flow showing the existing manual process and the operational problems it creates.

1.5.4 Future-State Business Flow (To-Be)

The target operating model uses a secure digital platform to standardize onboarding, assessment delivery, automated scoring, recommendation generation, certificate issuance, and policy-facing analytics.

Future-State Business Flow (To-Be)

The SDS platform standardises service delivery and produces operational and policy-ready outputs.

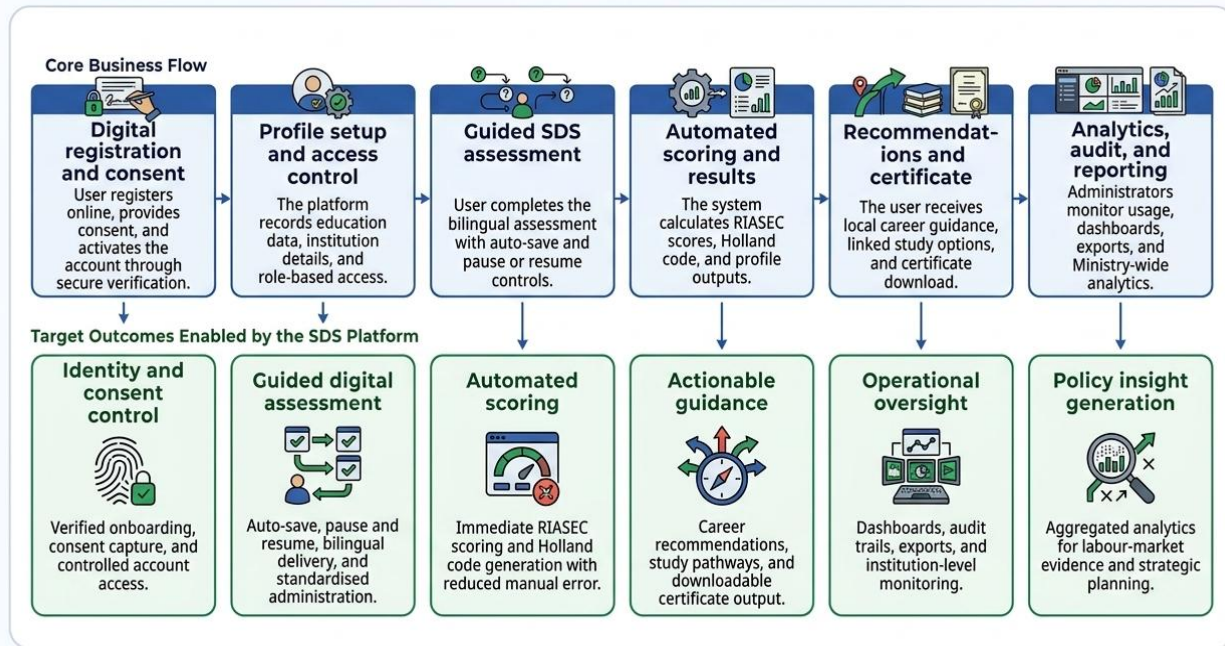


Figure 2. Future-state business flow showing the end-to-end digital process enabled by the SDS platform.

1.5.5 Value Proposition

Beneficiary	Value Delivered
For Users	Free, accessible, and scientifically validated career guidance.
For Institutions	Data-driven student counselling and programme alignment.
For the Ministry	Scalable services with auditable data and comprehensive analytics.
For Employers	A better career-aligned workforce development pipeline.

1.6 Product Value

The SDS Test System addresses priority career-guidance challenges in Eswatini by expanding access, standardizing service delivery, and producing evidence for operational and policy decisions.

1.6.1 Problem Solving

- **Accessibility gap:** provides services to underserved rural and urban populations.
- **Scalability limitation:** supports large-scale concurrent assessments beyond manual counselling capacity.
- **Data deficiency:** generates timely aggregated data for evidence-based planning.
- **Quality consistency:** standardizes assessment experience across regions and institutions.
- **Cost efficiency:** reduces effort and turnaround compared with manual processing.

1.6.2 Business Value Improvement

- Service coverage target: expand career guidance reach materially beyond current manual volumes.
- Processing time: reduce assessment-to-results time from weeks to minutes.
- Data accuracy: minimize manual capture and scoring errors through controlled workflows.
- Resource optimization: enables counsellors to focus on higher-value guidance activity.
- Policy insights: provide near-real-time evidence for planning and reporting.

1.6.3 Success Metrics

Metric	Target
User adoption	80% of secondary school students within two years of national rollout.
Assessment completion	90% completion rate for assessments that are started.
User satisfaction	4.5+ star average rating or approved equivalent satisfaction benchmark.
System uptime	99.5% availability during agreed business hours.
Data quality	95% or better accuracy in assessment and reporting outputs.

1.7 Motivation

The Ministry requires the SDS platform to address national priorities around youth employment, skills alignment, equitable access to guidance services, and digital transformation.

1.7.1 Strategic Alignment

- Supports national development and youth employment objectives.
- Aligns with education-sector modernization of career guidance services.
- Advances government digital service delivery and data-driven administration.
- Promotes equitable access to guidance services across the country.

1.7.2 Operational Needs

- Address shortage of qualified counsellors through scalable digital delivery.
- Provide consistent service quality across regions and institutions.
- Reduce operational cost and turnaround times.
- Enable immediate feedback to users and timely aggregated insight to the Ministry.
- Support integration with institutional and administrative processes.

1.7.3 Stakeholder Benefits

Stakeholder	Benefit
Students	Receive immediate, personalized career guidance.
Test Administrators	Gain tools for efficient student management and result review.
Institutions	Access evidence for curriculum and programme planning.
Policy Makers	Obtain comprehensive analytics and trends.
Employers	Benefit from better career-aligned workforce development.

1.8 Requirements Gathering Methodology

This SRS is based on stakeholder consultation, process review, prototype and feasibility validation, and compliance review. Reviewer comments received after issue of version 1.0 have been incorporated into this revision.

1.8.1 Information Collection Methods

- Stakeholder interviews with Ministry officials, counsellors, and institutional representatives.
- Group workshops with user groups including students, counsellors, and administrators.
- Questionnaires distributed to prospective users from different demographics.
- Review of current manual career guidelines and pain points.
- Observation of current assessment and counselling workflows.
- Analysis of existing career guidance materials, policies, and compliance obligations.

1.8.2 Validation Process

- Stakeholders review cycles with identified business owners and reviewers.
- Usability and prototype review with representative users.
- Technical feasibility validation of solution and deployment requirements.
- Legal and regulatory compliance review.
- Cost-benefit review against the intended operating model.

2. Overall Description

2.1 Product Perspective

- The SDS Test System is a web-based career assessment platform.
- It implements Holland's RIASEC personality assessment model.
- It provides career recommendations based on user responses and local pathways.
- It supports multiple user roles with differentiated permissions and workflows.
- It integrates with external email and file services and relies on PostgreSQL as the primary data store.
- It is intended for containerized deployment and production orchestration on Kubernetes.

2.2 Use Case Overview

2.2.1 System-Level Use Cases

Functional Area	Summary
User Management	Account registration, authentication, password reset, and profile management.
Assessment Administration	Delivery and tracking of the 228-question RIASEC assessment.
Results and Recommendations	Holland code generation, recommendations, and certificate output.
Analytics and Reporting	Dashboards, exports, and reports for authorized stakeholders.
Content Management	Questions, institutions, occupations, and managed content.
System Administration	Configuration, monitoring, audit, and operational controls.

2.2.2 Test Taker Use Cases

Primary: Register Account, Complete Assessment, View Results, Download Certificate, Manage Profile, Upload Qualifications.

Secondary: Reset Password, Update Profile, Export Personal Data, Delete Account.

2.2.3 Test Administrator Use Cases

Primary: Manage Students, Import Student Data, View Student Results, Generate Reports, Create Login Cards, Monitor Progress.

Secondary: Update Student Information, Delete Student Records, Export Student Data, View Institution Statistics.

2.2.4 System Administrator Use Cases

Primary: User Management, Role and Permission Management, Content Management, System Configuration, Analytics and Reporting, Audit Log Management.

Secondary: System Monitoring, Backup Management, Security Configuration, Performance Optimization, Integration Management, Deployment Oversight.

2.2.5 External System Interactions

External Interaction	Description
External Email Service	Bidirectional: verification messages, notifications, and password reset emails.
External File Storage	Document and certificate storage and retrieval.
PostgreSQL Database	Primary persistent data store for system entities and transactional records.
Report Generation	PDF and CSV export for certificates, reports, and data exports.
CI/CD Platform	Automated build, test, security scan, and staging deployment pipeline.
Infrastructure Platform	Container registry, Kubernetes runtime, and supporting cloud or on-premises services.

2.3 Product Functions

- User registration and authentication.
- SDS assessment administration and controlled answer capture.
- Automatic RIASEC scoring and Holland code generation.
- Career recommendations are linked to local institutions and pathways.
- Administrative management of users, content, and configuration.
- Reporting, dashboards, notifications, and certificate generation.
- Operational deployment, monitoring, backup, and audit support.

2.4 User Characteristics

2.4.1 Test Takers

User Group	Characteristics
High School Students	Ages 15–18; basic computer literacy; English or siSwati speakers.
University Students	Ages 18–25; moderate computer literacy; career planning focus.
Professionals	Ages 25+; seeking career guidance or career change; varied technical ability.

2.4.2 Test Administrators

School counsellors and career guidance professionals with moderate computer literacy. They are responsible for student management and assessment oversight. English proficiency is required and experience in career guidance is expected.

2.4.3 System Administrators

Ministry IT and administrative personnel with advanced computer literacy. They are responsible for configuration, user management, deployment oversight, monitoring, and maintenance.

2.5 Operating Environment

Environmental Element	Requirement
Web Browsers	Chrome 90+, Firefox 88+, Safari 14+, Edge 90+ or approved equivalents.
Internet Connection	Minimum 2 Mbps for intended operation; lower-bandwidth optimization required where possible.
Screen Resolution	Minimum 1024 × 768; recommended 1920 × 1080.
Operating Systems	Windows 10+, macOS 10.15+, Ubuntu 18.04+, iOS 13+, Android 10+.
Deployment Runtime	Container runtime with Kubernetes orchestration for production environments.

2.6 Design and Implementation Constraints

- Technology stack: React.js frontend, Node.js/Express backend, PostgreSQL database.
- Authentication model: JWT-based access control with refresh tokens stored in HTTP Only cookies.
- Data protection: compliance with the Eswatini Data Protection Act 2022.
- Accessibility: WCAG 2.1 AA compliance is mandatory.
- Security: OWASP-aligned controls are mandatory.
- Language support: English throughout the system.
- Deployment model: the solution shall support Docker-based containerization and Kubernetes-based orchestration for production use.

2.7 Required Delivery Artefacts

- Complete source code for frontend, backend, database migrations, and automated tests.
- Docker files and related containers build assets.
- Kubernetes manifests or Helm charts for deployable environments.
- Terraform templates or approved equivalent infrastructure-as-code artefacts.
- Automated CI/CD pipeline definitions for build, test, security scan, and staging deployment.
- Technical documentation, API documentation, user documentation, and operations runbooks.

3. Functional Requirements

3.1 User Registration and Authentication

3.1.1 User Registration

ID	Requirement
FR-001	Users shall register with national ID, email, and password.
FR-002	The system shall automatically extract date of birth and gender from the 13-digit national ID.
FR-003	Users must provide explicit consent for data processing before completing registration.
FR-004	Email verification shall be required before assessment access is granted.
FR-005	Duplicate national ID or email addresses shall be rejected with a clear error message.
FR-006	Passwords must meet minimum security requirements: 8+ characters, combination of letters and numbers.

3.1.2 User Authentication

ID	Requirement
FR-007	Users shall authenticate using email, username, or student number.
FR-008	The system shall implement JWT-based stateless authentication.
FR-009	Access tokens shall expire after 1 hour.
FR-010	Refresh tokens shall be stored in HTTP Only cookies for security.
FR-011	Failed login attempts shall trigger rate limiting after 5 consecutive failed attempts.
FR-012	Password reset functionality shall be available via email link.

3.1.3 Profile Management

ID	Requirement
FR-013	Users shall complete onboarding profile during first login.
FR-014	Profile shall include education level, institution, and occupation preferences.
FR-015	Users shall upload qualification documents with associated metadata.
FR-016	Profile information shall be editable with proper validation at all times.
FR-017	Users shall export all personal data in compliance with GDPR principles.
FR-018	Users shall delete their accounts subject to data retention policies.

3.2 SDS Assessment System

3.2.1 Assessment Administration

ID	Requirement
FR-019	Users shall start assessments with unique session identifiers.
FR-020	Assessments shall present 228 questions across 6 RIASEC sections.
FR-021	Questions shall be categorized by RIASEC type and section.
FR-022	Progress shall be saved automatically after each question response.
FR-023	Users shall be able to pause and resume assessments within a 30-day window.
FR-024	Assessment sessions shall time out after 2 hours of inactivity.

3.2.2 Question Management

ID	Requirement
FR-025	Questions shall be stored with RIASEC classification and section metadata.
FR-026	Administrators shall create, update, and delete questions.
FR-027	Questions shall support import/export via CSV format.
FR-028	Question order shall be randomized within sections.
FR-029	Questions shall display in the user's preferred language (English or siSwati).
FR-030	System shall validate question text and formatting before publication.

3.2.3 Answer Processing

ID	Requirement
FR-031	Before the start of a section, the users will see a narrative on how to answer that section. Then the users shall answer questions with YES / NO / UNCERTAIN options, depending on the section they're in.
FR-032	Answers shall be stored with timestamps and response times.
FR-033	The system shall calculate RIASEC scores automatically upon completion.
FR-034	The scoring algorithm shall follow Holland's RIASEC methodology precisely.
FR-035	Results shall include raw scores and normalized percentages for each RIASEC dimension.
FR-036	The system shall generate a 3-character Holland code from the three highest scoring dimensions.

3.3 Results and Recommendations

3.3.1 Result Generation

ID	Requirement
FR-037	The system shall generate results immediately upon assessment completion.
FR-038	Results shall include Holland code, dimension scores, and personality description.
FR-039	The system shall provide occupation recommendations based on the user's Holland code.
FR-040	Recommendations shall include local labour market demand and education requirements.
FR-041	System shall suggest relevant academic subjects aligned to the Holland code.
FR-042	Results shall be downloadable as professionally formatted PDF certificates.

3.3.2 Career Recommendations

ID	Requirement
FR-043	Occupation recommendations shall match the user's Holland code.
FR-044	Recommendations shall include local Eswatini labour market information.
FR-045	Education requirements shall be specified for each recommended occupation.
FR-046	The system shall link occupations to relevant courses and local institutions.
FR-047	Users shall filter recommendations by education level or region.
FR-048	Recommendations shall be saved to the user's profile for future reference.

3.3.3 Certificate Generation

ID	Requirement
FR-049	The system shall generate completion certificates with unique verifiable identifiers.
FR-050	Certificates shall include user name, Holland code, and completion date.
FR-051	Certificates shall be downloadable in PDF format.
FR-052	Certificate validity shall be verifiable via a unique code on the certificate.
FR-053	Administrators shall reissue certificates if needed.
FR-054	Certificate templates shall be customisable by administrators.

3.4 Administrative Functions

3.4.1 User Management

ID	Requirement
FR-055	Administrators shall view, create, update, and delete user accounts.
FR-056	User lists shall support filtering, searching, and pagination.
FR-057	Administrators shall bulk import users via CSV files.
FR-058	User accounts shall support activation and deactivation.
FR-059	Administrators shall assign roles and permissions to users.
FR-060	User activity shall be logged for audit purposes.

3.4.2 Content Management

ID	Requirement
FR-061	Administrators shall manage institutions and occupations in the system.
FR-062	Institutions shall include name, type, region, and contact information.
FR-063	Occupations shall include Holland codes and education requirements.
FR-064	Content shall support bulk import/export operations.
FR-065	Pending submissions shall require administrative approval before publication.
FR-066	Content changes shall be tracked with audit logs.

3.4.3 System Configuration

ID	Requirement
FR-067	Administrators shall configure system settings and parameters.
FR-068	The permission system shall support granular access control.
FR-069	The system shall support multiple administrative roles.
FR-070	Configuration changes shall require proper authorisation.
FR-071	System settings shall be backed up regularly.
FR-072	Administrative actions shall require confirmation for critical operations.

3.5 Analytics and Reporting

3.5.1 Data Analytics

ID	Requirement
FR-073	The system shall provide analytics dashboards for administrators.
FR-074	Analytics shall include assessment completion rates and trends.
FR-075	Holland code distribution shall be visualised in charts.
FR-076	Regional analytics shall show geographic distribution of assessments.
FR-077	Institution-specific analytics shall be available to test administrators.
FR-078	Analytics data shall be exportable in various formats.

3.5.2 Reporting Functions

ID	Requirement
FR-079	The system shall generate comprehensive reports for administrators.
FR-080	Reports shall include user demographics and assessment statistics.
FR-081	Custom reports shall be created with date ranges and filters.
FR-082	Reports shall be downloadable in PDF and CSV formats.
FR-083	Scheduled reports shall be generated automatically.
FR-084	Report access shall be controlled by user permissions.

3.6 Notification System

3.6.1 Email Notifications

ID	Requirement
FR-085	The system shall send email verification notifications upon registration.
FR-086	Password reset emails shall be sent upon user request.
FR-087	Assessment completion notifications shall be sent to users.
FR-088	Administrative notifications shall alert staff to important system events.
FR-089	Email templates shall be customisable by administrators.
FR-090	Email delivery status shall be tracked and logged.

3.6.2 In-App Notifications

ID	Requirement
FR-091	In-app notifications shall inform users of important events.
FR-092	Notifications shall support read/unread status tracking.
FR-093	Users shall mark notifications as read individually or in bulk.
FR-094	Notification preferences shall be configurable by users.
FR-095	The system shall limit notification history to prevent storage issues.
FR-096	Critical notifications shall require user acknowledgment.

4. Non-Functional Requirements

4.1 Performance Requirements

4.1.1 Response Time

ID	Requirement
NFR-001	API response times shall be under 2 seconds for 95% of requests.
NFR-002	Page load times shall be under 3 seconds on standard broadband connections.
NFR-003	Assessment question loading shall complete in under 1 second.
NFR-004	Report generation shall complete within 30 seconds.
NFR-005	Database queries shall be optimised for sub-second execution.

4.1.2 Throughput

ID	Requirement
NFR-006	The system shall support 500 concurrent users.
NFR-007	The system shall handle 1,000 assessment submissions per hour.
NFR-008	API shall support 100 requests per second per endpoint.
NFR-009	File uploads shall support 50 concurrent uploads.
NFR-010	Email sending shall support 1,000 emails per hour.

4.1.3 Resource Utilisation

ID	Requirement
NFR-011	CPU usage shall remain under 80% under normal load conditions.
NFR-012	Memory usage shall not exceed 4 GB per application instance.
NFR-013	Database connections shall be pooled and limited to 100 per instance.
NFR-014	File storage shall be optimised with compression where possible.
NFR-015	Caching shall reduce database load by at least 40%.

4.2 Security Requirements

4.2.1 Authentication Security

ID	Requirement
NFR-016	Passwords shall be hashed using encryption with a minimum of 10 rounds.
NFR-017	JWT tokens shall use the RS256 signing algorithm.
NFR-018	Access tokens shall expire after a maximum of 1 hour.
NFR-019	Refresh tokens shall be stored in httpOnly, secure cookies.
NFR-020	Multi-factor authentication shall be available for all administrator accounts.

4.2.2 Data Protection

ID	Requirement
NFR-021	All data in transit shall be encrypted using TLS 1.3.
NFR-022	Sensitive data at rest shall be encrypted using AES-256.
NFR-023	Personal data shall be anonymised after the applicable retention period.
NFR-024	Data access shall be logged with user identification.
NFR-025	Data backup shall be encrypted and stored securely.

4.2.3 Access Control

ID	Requirement
NFR-026	Role-based access control shall be enforced for all system resources.
NFR-027	Permission checks shall occur at both the endpoint and data level.
NFR-028	Session timeout shall occur after 30 minutes of inactivity.
NFR-029	IP-based rate limiting shall prevent abuse and brute-force attacks.
NFR-030	Administrative actions shall require additional authorisation for sensitive operations.

4.2.4 Compliance

ID	Requirement
NFR-031	The system shall comply with the Eswatini Data Protection Act 2022.
NFR-032	GDPR principles shall be implemented for data processing activities.
NFR-033	Data retention policies shall be configurable and automatically enforced.
NFR-034	Right to erasure shall be implemented with complete data removal capability.
NFR-035	Data processing consent shall be explicit, informed, and revocable by the user.

4.3 Usability Requirements

4.3.1 Accessibility

ID	Requirement
NFR-036	The system shall comply with WCAG 2.1 Level AA accessibility guidelines.
NFR-037	All functionality shall be accessible via keyboard navigation.
NFR-038	Screen reader compatibility shall be maintained throughout the system.
NFR-039	Colour contrast ratios shall meet a minimum 4.5:1 ratio for normal text.
NFR-040	Alternative text shall be provided for all meaningful images.

4.3.2 User Experience

ID	Requirement
NFR-041	User interface shall be intuitive and consistent across all pages.
NFR-042	Error messages shall be clear, specific, and actionable.
NFR-043	Help documentation shall be available for all major system functions.
NFR-044	Mobile-responsive design shall work on screen sizes of 320 px and above.

4.3.3 Learnability

ID	Requirement
NFR-046	New users shall be able to complete the assessment without prior training.
NFR-047	The onboarding process shall guide users through initial setup step by step.
NFR-048	Common tasks shall be accomplishable within 3 clicks from the main dashboard.
NFR-049	The system shall provide contextual help at points where users may need guidance.

ID	Requirement
NFR-050	User feedback shall be collected, reviewed, and addressed in subsequent releases.

4.4 Reliability Requirements

4.4.1 Availability

ID	Requirement
NFR-051	The system shall maintain 99.5% uptime during business hours.
NFR-052	Planned maintenance windows shall not exceed 4 hours per month.
NFR-053	Critical functions shall have automated failover capability.
NFR-054	Database backups shall be performed daily with integrity verification.
NFR-055	The system shall recover from failures within 5 minutes via automated procedures.

4.4.2 Data Integrity

ID	Requirement
NFR-056	Database transactions shall maintain full ACID properties.
NFR-057	Data validation shall prevent invalid state transitions.
NFR-058	Concurrent access shall be handled with proper locking mechanisms.
NFR-059	Data corruption shall be detected and corrected automatically where possible.
NFR-060	Audit logs shall be immutable and tamper-evident.

4.4.3 Error Handling

ID	Requirement
NFR-061	The system shall handle errors without data loss.
NFR-062	Error messages shall be logged with sufficient detail for diagnosis.
NFR-063	Users shall be informed of recoverable errors with actionable guidance.
NFR-064	The system shall maintain core functionality during partial failures.
NFR-065	Error recovery procedures shall be documented and tested regularly.

4.5 Maintainability Requirements

4.5.1 Code Quality

ID	Requirement
NFR-066	Code shall maintain a minimum of 80% automated test coverage.
NFR-067	Code shall follow established coding standards and style guides.
NFR-068	Documentation shall be maintained for all public APIs.
NFR-069	Code reviews shall be required for all changes before merging.
NFR-070	Technical debt shall be tracked and addressed in a structured manner.

4.5.2 System Architecture

ID	Requirement
NFR-071	The system shall use modular architecture to facilitate independent maintenance.
NFR-072	Components shall be loosely coupled and highly cohesive.
NFR-073	Configuration shall be externalised from the application code.
NFR-074	Logging shall be comprehensive, structured, and centrally aggregated.
NFR-075	The system shall support fully automated deployment pipelines.

4.5.3 Documentation

ID	Requirement
NFR-076	Technical documentation shall be kept current with each release.
NFR-077	User documentation shall be available, searchable, and role-specific.
NFR-078	API documentation shall be generated automatically from code annotations.
NFR-079	System architecture documentation shall be maintained and version-controlled.
NFR-080	Change logs shall be maintained for all system releases.

4.5.4 DevSecOps and Deployment Readiness

ID	Requirement
NFR-081	The production solution shall be deployable on Kubernetes using container images and declarative deployment manifests or Helm charts.
NFR-082	The Kubernetes deployment shall support horizontal auto-scaling based on defined CPU, memory, and/or application performance metrics.
NFR-083	The production platform shall support self-healing through automatic restart, rescheduling, and health-probe-based recovery of failed workloads.
NFR-084	Infrastructure required to build, deploy, secure, and operate the system shall be delivered as Terraform templates or an approved equivalent infrastructure-as-code implementation.
NFR-085	A fully automated CI/CD pipeline shall be delivered with the source code and shall execute builds, automated tests, security scanning, and deployment to a staging environment.

5. External Interface Requirements

5.1 User Interfaces

5.1.1 Web Interface

ID	Requirement
UIR-001	The system shall provide a fully responsive web interface.
UIR-002	Interface shall support all modern web browsers (Chrome, Firefox, Safari, Edge).
UIR-003	Design shall follow a consistent visual identity aligned with Ministry branding.
UIR-004	Navigation shall be intuitive and predictable for all user roles.
UIR-005	Forms shall provide real-time validation feedback to users.

5.1.2 Mobile Interface

ID	Requirement
UIR-006	Mobile interface shall be touch-optimised for smartphone and tablet use.
UIR-007	Critical functions shall work offline with synchronisation capability when reconnected.
UIR-008	Mobile performance shall be optimised for slower connections (2 Mbps minimum).
UIR-009	Interface shall adapt correctly to different screen orientations (portrait and landscape).
UIR-010	Mobile-specific features shall utilise native device capabilities where beneficial.

5.2 Software Interfaces

5.2.1 API Interfaces

ID	Requirement
API-001	RESTful API shall follow the OpenAPI specification.
API-002	API responses shall use a consistent JSON format.
API-003	API versioning shall be implemented via URL path (e.g., /api/v1/).
API-004	API documentation shall be comprehensive and interactive (e.g., Swagger UI).
API-005	API shall support rate limiting and token-based authentication.

5.2.2 Database Interfaces

ID	Requirement
DB-001	PostgreSQL 14+ shall be used as the primary database.
DB-002	Database connections shall be pooled and optimised for performance.
DB-003	Database schema shall support migrations and version control.
DB-004	Database queries shall be parameterised to prevent SQL injection.
DB-005	Database backup and recovery procedures shall be automated and tested.

5.3 Hardware Interfaces

5.3.1 Server Requirements

ID	Requirement
HW-001	Application server shall have a minimum of 4 GB RAM.
HW-002	Database server shall have a minimum of 8 GB RAM.
HW-003	Storage shall be SSD-based with a minimum of 256 GB capacity.
HW-004	Network bandwidth shall support a minimum 2 Mbps connectivity.
HW-005	Redundant power supplies shall be implemented for high availability.

5.3.2 Client Requirements

ID	Requirement
HW-006	Client devices shall have a minimum of 4 GB RAM.
HW-007	Display resolution shall support a minimum of 1024×768 pixels.
HW-008	Network connection shall support a minimum of 2 Mbps throughput.
HW-009	A modern, up-to-date web browser shall be installed on client devices.
HW-010	Touch interface shall be supported for mobile and tablet devices.

5.4 Communication Interfaces

5.4.1 Network Protocols

ID	Requirement
NET-001	HTTPS/TLS 1.3 shall be used for all client-server communications.
NET-002	WebSocket shall be used for real-time features such as live notifications.
NET-003	SMTP shall be used for all outbound email communications.
NET-004	IPv4 and IPv6 shall both be supported.
NET-005	Network timeouts shall be configured appropriately for each service type.

5.4.2 Data Formats

ID	Requirement
DATA-001	JSON shall be used for all API data exchange.
DATA-002	PDF shall be used for document and certificate generation.
DATA-003	CSV shall be supported for data import and export operations.
DATA-004	Images shall support JPEG, PNG, and WebP formats.
DATA-005	UTF-8 character encoding shall be used throughout the system.

6. System Features

6.1 Feature: User Authentication and Authorisation

Provides secure access control for all system users with role-based permissions and granular access control.

Functional Requirements covered: FR-001 through FR-018.

Client Acceptance Criteria

Measure	Target
Registration Success Rate	95% of registration attempts complete successfully without technical errors.
Login Performance	90% of successful logins complete within 2 seconds.
Security Compliance	Zero critical security vulnerabilities in formal penetration testing at acceptance.
User Satisfaction	4.5-star or approved equivalent rating for authentication experience.
Support Ticket Reduction	Material reduction in password-related support requests after rollout.

Technical Acceptance Criteria

- Users can register and authenticate successfully.
- Role-based access control functions correctly for all defined roles.
- Password security requirements are enforced at registration and change.
- Data privacy controls are implemented and auditable.
- Multi-factor authentication works correctly for administrator accounts.

Business Acceptance Criteria

- The system supports target user volumes.
- Authentication costs remain within the approved budget.
- Integration with Ministry user directories can be supported where approved.
- Compliance review passes without unresolved findings.

6.2 Feature: SDS Assessment Engine

Implements Holland's RIASEC assessment model with interactive questionnaire and automatic scoring.

Functional Requirements covered: FR-019 through FR-036.

Client Acceptance Criteria

Measure	Target
Assessment Completion Rate	90% of started assessments are completed.
User Experience Score	4.3-star or approved equivalent rating for the assessment interface.
Scoring Accuracy	100% correlation with the approved manual RIASEC scoring methodology.
Assessment Time	Average completion time under 45 minutes.
Mobile Compatibility	95% completion rate on supported mobile devices.

Technical Acceptance Criteria

- The 228-question assessment functions correctly across all RIASEC sections.
- The scoring algorithm produces accurate Holland code results.
- Progress tracking and resume functionality work within the required window.
- Assessment data is securely stored and processed with integrity.
- The system handles required concurrency targets.

Business Acceptance Criteria

- Assessment availability meets uptime targets.
- Results are produced within agreed turnaround windows.
- Both English and siSwati are supported.

6.3 Feature: Results and Career Recommendations

Generates personalised career recommendations based on assessment results with local labour-market integration.

Functional Requirements covered: FR-037 through FR-054.

Client Acceptance Criteria

Measure	Target
Result Accuracy	95% user satisfaction with career recommendations.
Certificate Quality	Professional certificates accepted by intended institutions.
Recommendation Relevance	80% of users find recommendations helpful and actionable.
Download Success	98% successful PDF download rate across supported devices.
Mobile Access	90% of users can access results on mobile devices.

Technical Acceptance Criteria

- Holland code calculation is accurate.
- Career recommendations are relevant, personalised, and localised to Eswatini.
- Certificate generation works with unique verifiable identifiers.
- PDF downloads are functional and properly formatted.
- Results are accessible within the agreed response time.

Business Acceptance Criteria

- Labour market data is refreshed according to the agreed operational cycle.
- Certificate verification is available to authorised stakeholders.
- Result archives meet approved retention requirements.

6.4 Feature: Administrative Management

Provides tools for user management, content management, configuration control, and auditability.

Functional Requirements covered: FR-055 through FR-072.

Client Acceptance Criteria

Measure	Target
Administrator Productivity	75% reduction in administrative task completion time compared with the manual baseline.
Data Accuracy	99% accuracy in approved bulk import operations.
System Usability	4.2-star or approved equivalent administrator rating.
Training Time	New administrators trained within 2 working days.
Error Reduction	90% reduction in data-entry errors compared with the manual process.

Technical Acceptance Criteria

- Administrative functions are accessible only to authorised users.
- Bulk operations work efficiently for the approved data volume.
- Content approval workflows function end to end.
- Configuration changes are applied properly and are auditable.
- Audit trails capture all relevant administrative actions.

Business Acceptance Criteria

- Operational costs are reduced relative to the manual baseline.
- The system supports multiple administrators concurrently.
- Backups and recovery procedures are tested on a defined schedule.
- Compliance reporting is automated where approved.

6.5 Feature: Analytics and Reporting

Delivers analytics and reporting capabilities for the Ministry and other authorised stakeholders.

Functional Requirements covered: FR-073 through FR-084.

Client Acceptance Criteria

Measure	Target
Report Accuracy	98% accuracy in generated reports.
Decision Support	85% of stakeholders find the reports actionable for decision-making.
Report Generation	Standard reports generated within 2 minutes.
Data Visualisation	4.4-star or approved equivalent dashboard usability score.
Export Reliability	99% successful export rate across supported formats.

Technical Acceptance Criteria

- Dashboards display accurate and current data.
- Reports generated with correct formatting for supported formats.
- Scheduled reports are delivered on time to configured recipients.
- The system handles analytical workloads over large datasets.

Business Acceptance Criteria

- Reports support Ministry policy decisions with evidence-based insight.
- Monthly stakeholder reporting can be automated.
- Historical data is retained according to approved policy.

6.6 Feature: Notification System

Manages email and in-app notifications for user communication and operational alerting.

Functional Requirements covered: FR-085 through FR-096.

Client Acceptance Criteria

Measure	Target
Email Delivery	98% successful email delivery rate.
Notification Relevance	90% user satisfaction with notification content.
Response Time	Critical notifications delivered within 5 minutes.
Unsubscribe Rate	Under 5% unsubscribe rate from non-mandatory notifications.
Mobile Support	95% notification accessibility on supported mobile devices.

Technical Acceptance Criteria

- Email notifications are sent and received for defined triggers.
- In-app notifications display correctly across supported browsers.
- Notification preferences are respected.
- Notification history is maintained within configured storage limits.

Business Acceptance Criteria

- Notification costs remain within the approved budget.
- Templates are customisable by authorised administrators.
- English and siSwati support is available for notification content.

7. User Characteristics

7.1 Test Taker Profile

Characteristic	Description
Age Range	15–60 years.
Education Level	Varies from secondary to postgraduate.
Technical Proficiency	Basic to intermediate computer skills.
Language	English or siSwati speakers.
Geographic Location	Primarily Eswatini residents.

7.1.2 User Goals

- Identify suitable career paths based on interests and strengths.
- Understand education requirements for chosen careers.
- Access local labour-market information.
- Receive personalised, actionable guidance securely.

7.1.3 User Expectations

- Simple and intuitive assessment process requiring no prior training.
- Immediate results and recommendations after completion.
- Confidential and secure handling of personal information.
- Access from multiple device types, including smartphones.

7.2 Test Administrator Profile

Characteristic	Description
Professional Role	School counsellors and career advisors.
Education Level	Bachelor's degree or higher or equivalent approved experience.
Technical Proficiency	Intermediate computer skills.
Experience	2+ years in career guidance or related administration.
Language	English proficiency required.

7.2.2 User Goals

- Manage student assessment processes efficiently.
- Monitor progress and review results.
- Generate reports for school leadership and Ministry stakeholders.
- Import and manage bulk student data.

7.2.3 User Expectations

- Efficient bulk operations at scale.
- Comprehensive reporting and export capability.
- Reliable performance during peak assessment periods.

- Secure handling of student data in compliance with regulation.

7.3 System Administrator Profile

Characteristic	Description
Professional Role	IT staff and Ministry administrators.
Education Level	Technical degree, certification, or approved equivalent experience.
Technical Proficiency	Advanced computer skills.
Experience	3+ years in system administration or support.
Language	English proficiency required.

7.3.2 User Goals

- Maintain system security, integrity, and performance.
- Manage user accounts, roles, and permissions.
- Configure system settings and maintain content.
- Monitor system health, usage, deployment, and compliance.

7.3.3 User Expectations

- Comprehensive administrative tools with a full audit trail.
- Detailed monitoring dashboards and alerting.
- Reliable backup and disaster recovery capability.
- Clear technical and operational documentation.

8. Constraints

8.1 Technical Constraints

8.1.1 Technology Stack

Area	Constraint
Frontend	React.js 18+ with modern JavaScript features.
Backend	Node.js 18+ with Express.js framework.
Database	PostgreSQL 14+ with controlled schema management.
Authentication	JWT-based authentication with refresh tokens.
Deployment	Must support containerised Docker deployment and production Kubernetes orchestration.

8.1.2 Performance Constraints

- API responses must meet the response-time requirements defined in Section 4.
- The system must support the defined concurrency and throughput targets.
- The database design must scale to large data volumes.
- File storage and exports are limited by approved infrastructure capacity.
- The solution must remain usable on the minimum network conditions defined in this SRS.

8.1.3 Security Constraints

- Compliance with the Eswatini Data Protection Act 2022 is mandatory.
- TLS 1.3 is required for all network communication.
- Multi-factor authentication is mandatory for administrator accounts.
- A complete, tamper-evident audit trail is required.
- Role-based access control is mandatory for all protected resources.

8.1.4 Delivery Constraints

- Production deployment shall be designed for Kubernetes-based orchestration.
- Infrastructure shall be codified and delivered as Terraform templates or approved equivalent.
- Automated CI/CD definitions shall form part of the delivered source code.
- Third-party components shall not introduce copyleft or other ownership restrictions prohibited by the Ministry.

8.2 Business Constraints

8.2.1 Budget Constraints

- Development is subject to an approved budget and managed scope.
- Infrastructure costs must align to approved government hosting capacity.
- Maintenance and support budgets are constrained.
- Training budgets are limited and must be used efficiently.

8.2.2 Time Constraints

- Project implementation is expected within the approved contractual period.
- Formal testing windows are limited and must be planned carefully.
- Training must be completed before the approved go-live date.
- Data migration, where applicable, must fit the approved migration window.

8.2.3 Regulatory Constraints

- Data protection and privacy requirements are mandatory.
- WCAG 2.1 AA compliance is mandatory.
- Government reporting and audit obligations must be satisfied.

8.3 Operational Constraints

8.3.1 Staffing Constraints

- Internal technical resources at the Ministry are limited.
- Administrative resources may be part-time.
- User support capacity is limited and requires efficient tooling.
- Maintenance responsibilities may be shared across roles.

8.3.2 Infrastructure Constraints

- Government hosting infrastructure has defined limitations.
- Bandwidth is limited in some geographic regions.
- Existing hardware should be used where feasible and approved.
- Backup and monitoring tooling may be constrained by existing estate capability.

8.3.3 Process Constraints

- Government procurement processes apply to purchases and major changes.
- Formal change approval is required for approved-scope changes.
- Extensive documentation and formal testing procedures apply.
- A staged deployment approach is required by policy.

9. Assumptions and Dependencies

9.1 Assumptions

9.1.1 Technical Assumptions

- Users have access to supported devices and browsers.
- Users have sufficient internet access at the point of assessment.
- Users possess at least basic digital literacy.
- Users can read English or siSwati at a suitable level.

9.1.2 Business Assumptions

- Target users will adopt the system with proper communication and training.
- Input data provided by institutions will be reasonably accurate.
- Ministry leadership will continue to sponsor and support implementation.
- Funding will remain available through the agreed delivery lifecycle.

9.1.3 Operational Assumptions

- Scheduled maintenance will be performed as agreed.
- Backup procedures will be followed consistently.
- Security practices and policies will be maintained.
- Support resources will be made available to users.

9.2 Dependencies

Dependency	Purpose
Email Service	SMTP or approved email service for outbound notifications and resets.
Certificate Authority	Digital certificates for HTTPS/TLS encryption.
DNS Service	Domain name resolution for the system URL.
Time Service	Reliable NTP or equivalent synchronisation service.
Backup Service	External or approved backup storage service for offsite protection.
Container Registry	Storage and retrieval of deployable container images.
Kubernetes Platform	Production orchestration environment for workloads and scaling.
CI/CD Platform	Pipeline runtime for build, test, security scan, and staging deployment.

9.3 Risks

Risk	Impact	Mitigation
Performance does not meet requirements	High	Load testing in pre-production and performance optimization.
System does not scale as expected	High	Design for horizontal scaling from the outset; validate on Kubernetes.
Security vulnerabilities discovered	Critical	Formal security testing and OWASP-aligned controls.

Risk	Impact	Mitigation
Browser compatibility issues	Medium	Automated cross-browser test execution in CI/CD.
Data loss or corruption	Critical	Daily backups and restoration testing.
Project delay	Medium	Managed sprints, formal progress tracking, and change control.
Low user adoption	High	Awareness, training, and phased rollout support.

10. Requirements Traceability

10.1 Functional Requirements Traceability Matrix

Req. ID	Feature	Priority	Status	Test Case	Verification
FR-001	User Registration	High	Implemented / To Verify	TC-001	Automated
FR-004	Email Verification	High	Implemented / To Verify	TC-004	Automated
FR-020	228 Questions	High	Implemented / To Verify	TC-020	Automated
FR-033	RIASEC Scoring	High	Implemented / To Verify	TC-033	Automated
FR-037	Result Generation	High	Implemented / To Verify	TC-037	Automated
FR-049	Certificate Generation	High	Implemented / To Verify	TC-049	Automated
FR-055	User Management	High	Implemented / To Verify	TC-055	Manual
FR-073	Analytics Dashboard	Medium	Implemented / To Verify	TC-073	Manual
FR-085	Email Notifications	Medium	Implemented / To Verify	TC-085	Automated

10.2 Non-Functional Requirements Traceability

Req. ID	Category	Priority	Status	Test Case	Verification
NFR-001	Performance	High	Implemented / To Verify	TC-NF-001	Perf. Test
NFR-016	Security	High	Implemented / To Verify	TC-NF-016	Security Test
NFR-021	Data Protection	High	Implemented / To Verify	TC-NF-021	Security Test
NFR-031	Compliance	High	Implemented / To Verify	TC-NF-031	Review
NFR-036	Accessibility	High	Implemented / To Verify	TC-NF-036	WCAG Audit
NFR-051	Availability	High	Implemented / To Verify	TC-NF-051	Monitoring

Req. ID	Category	Priority	Status	Test Case	Verification
NFR-061	Error Handling	High	Implemented / To Verify	TC-NF-061	Negative Test
NFR-081	Kubernetes Deployment	High	New / To Verify	TC-NF-081	Review / Demo
NFR-082	Auto-scaling	High	New / To Verify	TC-NF-082	Perf. Test
NFR-083	Self-healing	High	New / To Verify	TC-NF-083	Chaos / Recovery Test
NFR-084	Infrastructure as Code	High	New / To Verify	TC-NF-084	Artefact Review
NFR-085	Automated CI/CD	High	New / To Verify	TC-NF-085	Pipeline Demo

10.3 User Acceptance Testing (UAT)

UAT shall confirm that the delivered solution satisfies approved business requirements and is suitable for operational deployment. UAT shall be executed against an agreed staging environment and approved test scenarios. The defect severity matrix below defines acceptance thresholds and remediation timelines.

Severity	Definition	Examples	Remediation Timeline	UAT Disposition
Critical	A defect that prevents use of a core business function, causes data loss or corruption, creates a material security or compliance exposure, or blocks go-live.	System unavailable; assessment cannot be submitted; unauthorised access to personal data; incorrect scoring affecting result integrity.	Fix or approved workaround required within 1 business day.	No open Critical defects are permitted for UAT sign-off.
Major	A defect that materially degrades a significant function, produces incorrect non-critical output, or causes repeated user failure without a reasonable workaround.	Certificate generation fails for some users; reporting exports incorrect fields; bilingual display broken on a key workflow.	Fix or approved workaround required within 3 business days.	Open Major defects may only be accepted if explicitly approved by the Ministry and do not prevent safe operation.
Minor	A defect with limited business impact, cosmetic issues, low-risk usability issues, or issues with a clear workaround that do not block core service delivery.	Minor layout issues; wording corrections; non-critical formatting defects; small dashboard presentation issues.	Fix in the next planned release or within 10 business days, unless otherwise agreed.	Minor defects may remain open at sign-off if logged, agreed, and scheduled for resolution.

10.3.1 UAT Exit Criteria

- All planned UAT scenarios have been executed and recorded.
- No Critical defects remain open.
- All Major defects have been resolved, formally waived, or have approved workarounds accepted by the Ministry.
- Residual Minor defects are logged, prioritised, and accepted by the Ministry.
- Required evidence, including test logs, screenshots, pipeline evidence, and deployment artefacts, have been submitted.

10.4 User Story Mapping

10.4.1 Test Taker User Stories

- As a test taker, I want to register with my national ID so that my profile is automatically created with demographic information.
- As a test taker, I want to complete the SDS assessment so that I can discover my career interests and strengths.
- As a test taker, I want to receive personalized recommendations so that I can make informed decisions about my future.

10.4.2 Test Administrator User Stories

- As a test administrator, I want to import student data in bulk so that I can manage multiple students efficiently.
- As a test administrator, I want to view student results so that I can provide informed individual guidance.
- As a test administrator, I want to generate reports so that I can share insight with school leadership and Ministry stakeholders.

10.4.3 System Administrator User Stories

- As a system administrator, I want to manage user accounts and permissions so that I can control access and maintain security.
- As a system administrator, I want to configure settings and deployment controls so that I can maintain reliable operation.
- As a system administrator, I want to monitor system health and recovery behaviour so that I can respond proactively to issues.

11. Appendix

11.1 Glossary

Term	Definition
SDS	Self-Directed Search — Holland career assessment tool.
RIASEC	Holland's six personality types: Realistic, Investigative, Artistic, Social, Enterprising, Conventional.
Holland Code	Three-character code derived from the highest RIASEC dimension scores.
JWT	JSON Web Token for secure stateless authentication.
WCAG	Web Content Accessibility Guidelines.
RBAC	Role-Based Access Control.
API	Application Programming Interface.
TLS	Transport Layer Security for data in transit.
ACID	Atomicity, Consistency, Isolation, Durability transaction properties.
CI/CD	Continuous Integration and Continuous Delivery/Deployment pipeline.
IaC	Infrastructure as Code.
Terraform	Infrastructure-as-code tooling used to define reproducible infrastructure.
Kubernetes	Container orchestration platform supporting scheduling, scaling, and self-healing.

11.2 References

- Holland, J. L. (1997). Making Vocational Choices: A Theory of Vocational Personalities and Work Environments.
- OWASP Top 10 2025 — Web Application Security Risks.
- WCAG 2.1 Accessibility Guidelines — W3C Recommendation.
- PostgreSQL Documentation v14.
- React Documentation v18.
- Eswatini Data Protection Act 2022.
- System Design Document (SDS Test System).

11.3 Intellectual Ownership

All source code, documentation, database schemas, infrastructure templates, deployment scripts, test artefacts, diagrams, and all other deliverables produced for the SDS Test System shall be the sole property of the Ministry of Labour and Social Security upon delivery and acceptance in accordance with the governing contract.

The delivered solution shall not contain third-party components, dependencies, or licensing terms that impose copyleft obligations or otherwise restrict the Ministry's right to use, host, modify, maintain, transfer

within government, or enhance the system. Any approved third-party components shall be fully disclosed and shall be subject only to licensing terms acceptable to the Ministry.

11.4 Document Approval

Role	Name	Signature	Date
Project Sponsor			
System Owner			
Technical Lead			
Quality Assurance			
Business Analyst			

11.5 Document Control

Item	Value
Classification	Confidential
Distribution	Ministry of Labour and Social Security, implementation team, approved stakeholders
Review Schedule	Quarterly
Next Review Date	July 2026
Document Owner	Measurement and Testing Unit, Ministry of Labour and Social Security (System Owner)
Prepared By	SDS Development Team
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